

Case 2: SONIA

Market: SONIA Z5 | Reference Level: 96.16

Part 1: The Quantitative Framework (My Method)

To solve this, I built a Python-based Scenario-Weighted Valuation Engine using the Bachelier Model, which is the standard for STIR options.

1. Calibration (The Input)

- **Challenge:** The case gave conflicting signals (a 'Sleeping' 25bps market vs. an 'Active' higher vol view anticipating the cut). Using either one blindly would create bad results.
- **My Solution:** I wrote a calibration script using `brentq` to solve for the Implied Volatility that matches the market price of the benchmark strategy (X5 96.25/96.35 Call Spread @ 1.25 ticks). I deliberately chose Local Calibration over Global Optimization. The examples provided included both Call Spreads (Directional) and Call Flies (Vol/Skew). In STIR markets, Skew is significant. A simple flat-volatility model cannot perfectly match a Call Spread and a Call Fly simultaneously because they imply different parts of the volatility surface. Since the client's trade is a Directional Breakout, the X5 Call Spread is the most relevant 'Liquid Anchor.' It represents the market's cleaner view on the move from 96.25 to 96.35. If I had averaged the volatility across all examples, I would have diluted the signal. I preferred to price the breakout correctly and accept some error on the flies, rather than pricing everything 'mediocrely'.
- **Result:** The model calibrated to **~44.4 bps**. This ensures my pricing is grounded in *reality*, not fantasy.

2. The Scenario Engine (The Logic)

- **Challenge:** The client has a view "Through December." A simple leverage calculation (Payoff/Cost) unfairly favors November (X5) options because they are cheaper, ignoring the risk that they expire *before* the cut.
- **My Solution:** I implemented a Weighted Scenario Class in Python:

From the view of the client I made the assumption that:

1. **Scenario A (Early Cut):** 20% Probability.
 2. **Scenario B (Late Cut):** 80% Probability.
 3. **Logic:** If the strategy is X5, $\text{Payoff_B} = 0$.
- **Impact:** This penalized the "cheap" November options and forced the model to value the *certainty* of the December (Z5) contract.

3. The "Dynamic" Filter

- **Challenge:** The client asked for "Dynamic" (Long Delta/Gamma). Standard Call Spreads become *Short Gamma* as they profit.
- **My Solution:** I added a 'Gamma Check' to the evaluation loop.
 - $\text{Gamma_Target} > 0 = \text{Dynamic}$ (Passes the filter).
 - $\text{Gamma_Target} < 0 = \text{Static}$ (Flagged as 'Capped').

Part 2: The Trade Recommendation (The Results)

Based on this calibrated engine, the model produced a clear hierarchy.

The Winner: Z5 Ratio 1x2 (Sell 96.15 / Buy 2x 96.25)

- **Score:** Highest Weighted EV/Cost (6.0).
- **Why it wins:**
 1. **Dynamic:** It is structurally Long Gamma. As the rate cut deepens, the position size *doubles*.
 2. **Efficient:** It funds the long volatility (OTM calls) by selling the expensive ATM strike (96.15).
 3. **Timing:** It uses the Z5 expiry to guarantee coverage of the December meeting.

The Hedge/Alternative: Z5 Call Spread (96.30 / 96.40)

- **Role:** The "Conservative" Choice.
- **Why:** If the client wants to target a *precise* landing (96.35) and is risk-averse, this maximizes absolute cash PnL. It sacrifices Gamma for Certainty.

Part 3: Risk Analysis (The "Valley of Death")

I must highlight the specific risk of the theoretical Winning Strategy (Ratio 1x2).

The Ratio structure has a 'dip' in the PnL if the market rallies slightly (to 96.25) and stops. If there is a 'weak' cut, you lose on the short leg (96.15) but don't make enough on the long legs (96.25) to cover it.

This strategy is a 'conviction trade.' If the client is unsure about the *magnitude* of the cut, I would recommend the Call Spread instead.

Part 4: Questions

Q1: How can two expirations be compared rationally in this context?

To compare X5 (November) and Z5 (December) rationally, we must move beyond simple Leverage (Payoff / Cost) and use Probability-Weighted Expected Value based on the Central Bank's calendar. If we compare them purely on price, X5 will always appear superior. It has less time value, so it is cheaper. This is irrational because it ignores the fact that X5 expires November 14, while the key event ("One cut through December") likely occurs at the December 18 meeting.

The Solution:

We incorporate the Probability of Survival, we assign a probability weight to the cut happening *before* November 14 (e.g., 20%) versus *after* November 14 (e.g., 80%).

By normalizing for "Event Capture Probability," Z5 emerges as the rationally superior trade despite the higher upfront cost. We are paying an extra premium to insure against the risk of the cut happening late.

Q2: How can the “dynamic” nature of a strategy be measured objectively?

We measure "dynamic" behavior by explicitly calculating the Net Gamma at the Target Strike (96.35).

We evaluate the Gamma of the total structure when the Spot Price equals the Target Price.

- **If Gamma > 0:** The strategy is Dynamic. As the market moves in your favor, your position gets *larger* (Delta increases). You are accelerating into the profit.
- **If Gamma < 0:** The strategy is Static/Capped. As the market moves in your favor, your position gets *smaller* (Delta decreases). You are decelerating or "fighting" the move.

Call Spreads (e.g., 96.30/96.40): At the target (96.35), the short 96.40 leg dominates the profile, turning the Net Gamma negative. This mathematically proves the strategy is "turning off" just as the trend accelerates. It fails the dynamic test.

Ratio 1x2 (Sell 1 / Buy 2): At the target, the two long calls overpower the single short call, keeping Net Gamma positive. This objectively measures that the strategy is participating more aggressively as the breakout occurs.

Q3: How can theta cost and roll cost be incorporated into the comparison?

We incorporate these costs by shifting our evaluation metric from "Return on Capital" (Payoff / Entry Price) to "Return on Carry" and "Event-Weighted Validity."

1. Incorporating Theta Cost (The "Rent" Adjustment)

Since the client's view is "Theta Negative" (long option), looking only at the entry price is deceptive. A cheap option that decays rapidly is expensive to hold.

We explicitly calculate the Net Daily Theta for each strategy and compare it against the Gamma gained, we prioritize strategies that finance their own decay:

Standard Call Spread: Pays Net Theta (You own the lower strike, which has higher theta).

Ratio 1x2: Neutral/Positive Theta. By selling the expensive ATM option (96.15), the strategy collects enough premium to offset the cost of the OTM "lottery tickets" (96.25s).

In the comparison, we penalize strategies with high daily bleed (pure long calls) and reward strategies that minimize "Rent" (Ratios), even if their upfront cost appears similar.

2. Incorporating Roll Cost (The "Binary" Filter)

"Roll Cost" in this context refers to the risk of the option expiring *before* the market move occurs, forcing the trader to pay a second premium to re-enter the trade.

Instead of trying to guess the price of the roll, we modeled this structurally using Probability-Weighted Scenarios.

Scenario: The rate cut happens "Through December".

- **X5 Expiry (Nov 14):** Has a high probability (I estimated 80%) of expiring *before* the cut.
- **Z5 Expiry (Dec 12):** Covers the full window.

We assigned a Payoff of Zero to X5 strategies in the "December Cut" scenario.

This mathematically treats the "Roll Cost" of X5 as 100% of the premium (Total Loss). This "Infinite Roll Penalty" in the model objectively disqualifies the November expiry without needing subjective guessing.

Q4: How can the EURIBOR case model be adapted to a long volatility / directional view?

The EURIBOR model was designed for Short Volatility, Long Theta. The SONIA model must solve for Breakout (Long Volatility, Negative Theta).

We stopped "fitting the skew" (looking for expensive wings/flies) and started "fitting the trend". We calibrated our volatility to the price of the ATM Call Spread, ensuring the model respects the cost of the breakout rather than the cost of the tails.

Adaptation 2 (Payoff Function): The EURIBOR model optimized for a "peak" profit at a specific number (98.06). The SONIA model was adapted to optimize for Slope (Delta/Gamma). We replaced the "Target Zone" logic with a "Threshold" logic once the price clears 96.35, the strategy with the steepest PnL slope (the Ratio) wins.

Q5: Which criteria should change in weight, and why?

Gamma (Weight: Low => Critical):

In EURIBOR, you want stability, so Gamma is a risk to be minimized. In SONIA, the client specifically asked for a "Dynamic" strategy. We increased the weight of Gamma from "Check" to "Driver." This is why the Ratio 1x2 (Gamma > 0) beat the Call Spread (Gamma < 0), even though the Call Spread is safer.

Timing/Expiry (Weight: Standard => Binary):

In EURIBOR, the view was valid "until Sep 2026". In SONIA, the view is tied to a specific central bank meeting cycle ("One cut through December"). We changed the weighting of Expiry from a continuous variable (time value) to a Binary Constraint (Must cover December). This disqualifies X5 despite its cheapness.

Cost Definition (Weight: "Cheapest Entry" => "Efficiency"):

In EURIBOR, you want "The cheapest strategy at entry" because your upside is capped. In SONIA, we deprioritized raw "Cheapness" in favor of "Leverage Efficiency" (Payoff vs. Event Probability). We are willing to pay *more* (for Z5) to ensure we actually catch the move.